MACSE622 – DATA ENGINEERING AND VISUALIZATION LAB

Register Number	25MCB1010	Subject Code	MACSE622
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Programme	MTech CSE (Big Data Analytics)	Slot	L33 + L34
Course	Data Engineering and Visualization Lab	Semester	2nd Semester
Date	19-01-2026	Exp No	5

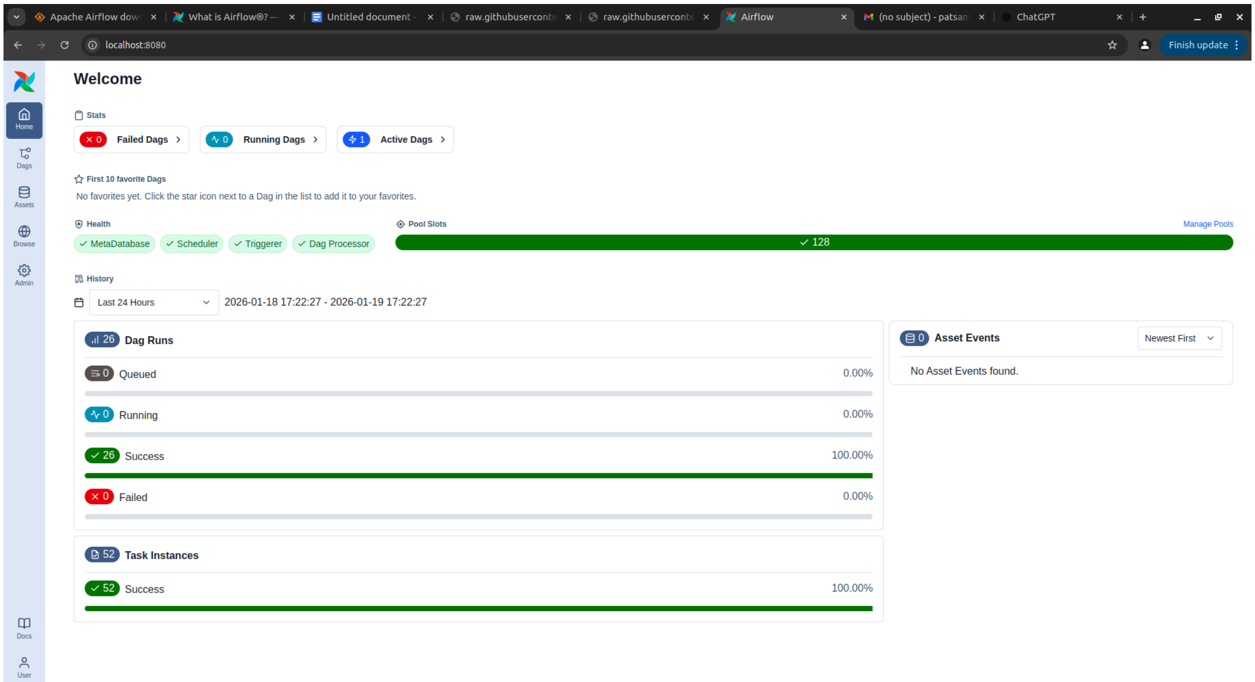
Ex No: 5.

Aim: Program an Apache Airflow ETL

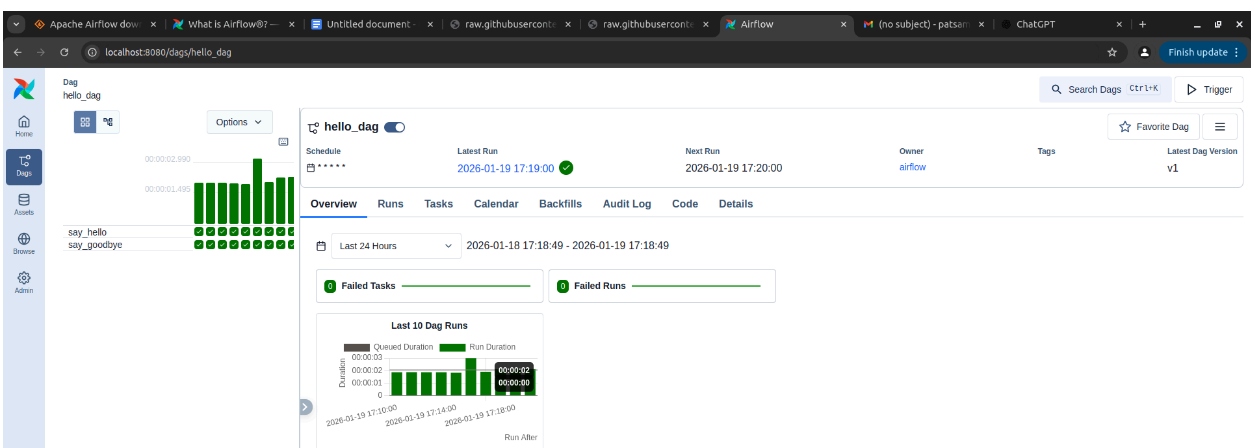
Implementation:

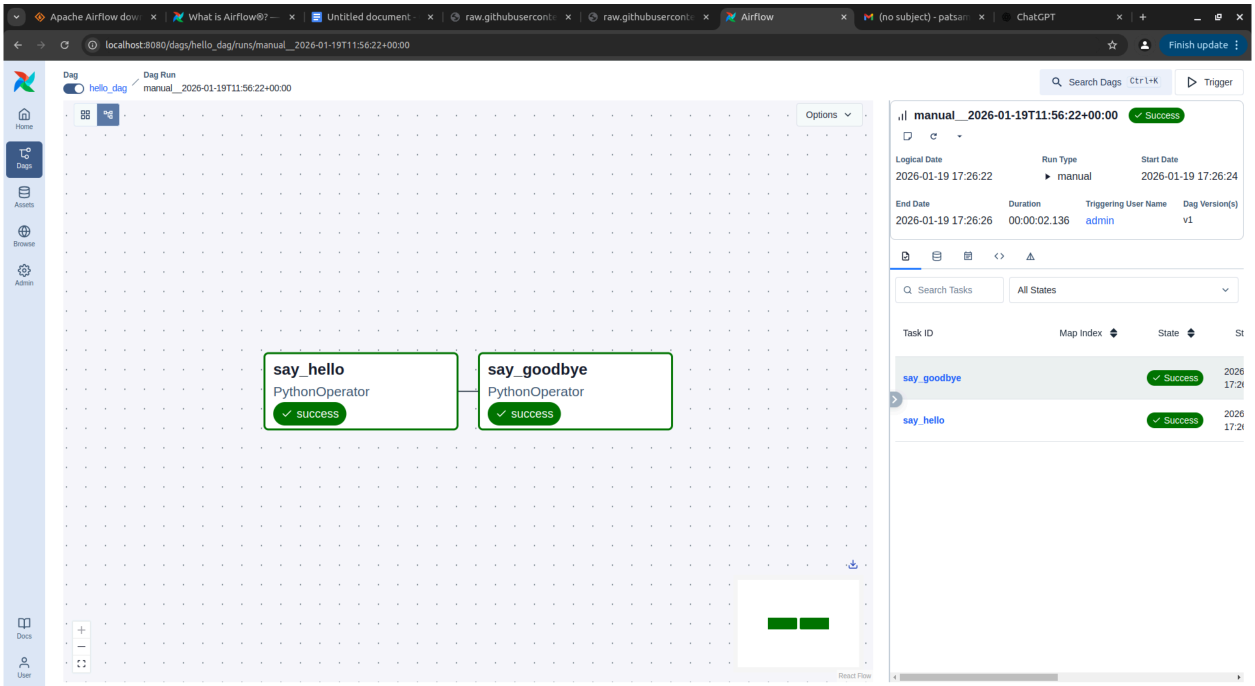
1. Environment Setup:
    a. Open a new terminal
    b. Go to home directory : **~cd**
    c. Install Python venv package : **sudo apt install python3.10-venv**
2. Virtual Environment
    a. Create venv : **python3 -m venv airflow_venv**
    b. Verify the folder exists : **ls ~/airflow_venv**
3. Activate venv
    a. Activate venv : **source ~/airflow_venv/bin/activate**
    b. Ensure the prompt shows **airflow_venv**
4. Install Airflow
    a. Run pip install with constraints for version compatibility
5. Initialise database
    a. Using path : **airflow db migrate**
6. Start Standalone
    a. Using command : **airflow standalone**
    b. Starts the scheduler + api + web Ui start
    c. Access via http://localhost:8080
7. Create DAG file
    a. Navigate to : **~/airflow/dags/**

    b. Create python file : **nano hello_dag.py**
    c. The code is attached in the document
    d. Save and exit : (Ctrl+O, Enter, Ctrl+X in nano)
8. Check DAG in UI
    a. Login with the default admin credentials in http://localhost:8080
    b. Verify if hello_dag DAG appears



9. Trigger DAG
    a. Manually trigger or wait for schedule
    b. Navigate through View logs -> see “Hello…” and “Goodbye..” messages

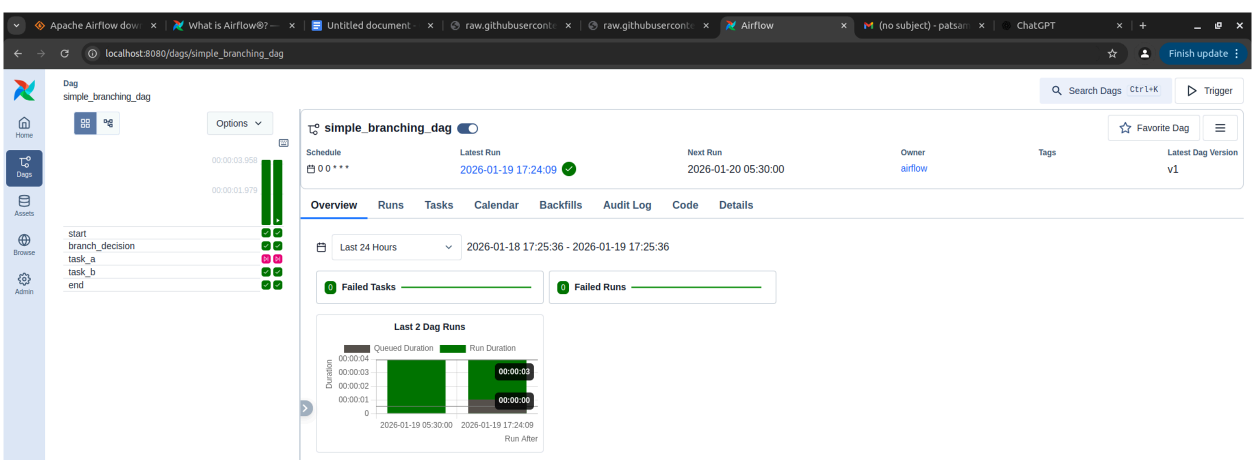


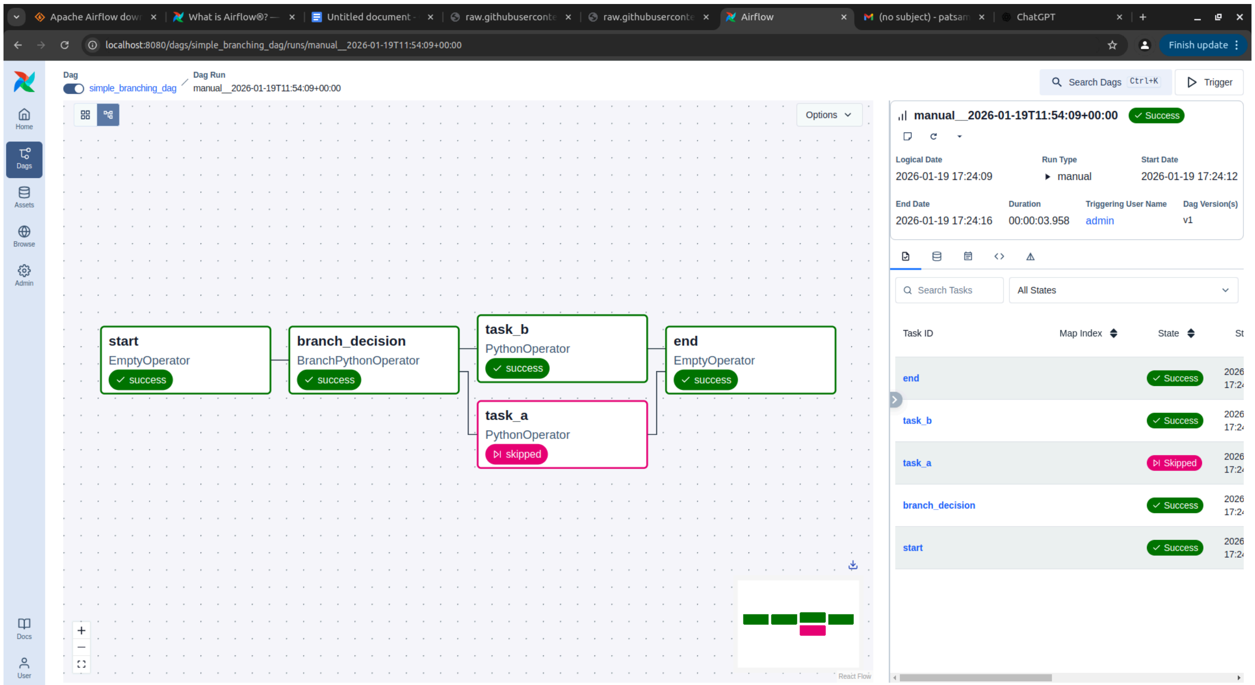


Explanation of **Program 1: hello_dag.py**

1. This code defines a DAG in Apache Airflow called hello_dag. The DAG runs every minute starting from Jan 12, 2026.
2. It contains two Python tasks: one prints “Hello from Airflow 3.x DAG!” and the other prints “Goodbye from Airflow DAG!”.
3. The tasks are connected so that the hello task always runs before the goodbye task.
4. The expected output is that each run will first show the hello message in the logs and then the goodbye message.

Open a new terminal window. Perform the above operations for **2nd program on branch DAG** simple_branching_dag.py





Explanation of **Program 2 : simple_branching_dag.py**

1. This code defines a DAG in Apache Airflow called simple_branching_dag. The DAG runs once every day starting from January 19, 2026.
2. It begins with a start task that does nothing.
3. A branching task then randomly chooses between two paths: task A or task B.
4. Task A prints “Running Task A” and Task B prints “Running Task B”.
5. Only one of these tasks will run each day, depending on the random choice.
6. The DAG ends with a final task that completes successfully if at least one branch has run.
7. The expected output is that the logs will show either the Task A message or the Task B message, but not both.

The codes is attached in the assignment folder in the under the name

1. hello_dag.py
2. simple_branching_dag.py

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